

How are we to live?

Ages 16+

*Evidence-based pointers on living well —
each with how it was tested, how big the effect is, and its
weakest point.*

Trust but verify

The method, stated plainly. Every claim on this site is presented with three things attached: its **study design** (how the evidence was gathered — an experiment, or following people over time), its **effect size** (how big the difference is), and its **honest weakness**. Every figure is tracked in a public **provenance** ledger, checked against its source and linked so you can go and read the original for yourself. This is a reference, not a campaign: no calls to action, no donate button. If it ever reads as a nice-looking set of assertions, it has failed — a confident claim with no working shown is no better than the marketing it hopes to improve on.

Trust but verify (cont.)

"Trust but verify" also sets the tone for how these topics are pitched at each age — reassurance where the real risk is low, straight talk where it isn't, and never a scary number without the honest context around it.

And here is the honest boundary of the whole enterprise, put where you can see it:

Trust but verify (cont.)

This entire site assumes you **care what happens to a future version of you** — the you of ten, forty, seventy years from now. Almost every argument here is a trade: a small cost now for a larger benefit to that future person. If you genuinely do not care what happens to future-you, and you are *certain* your future self will never feel differently, then there is nothing for you here — the trade has no value if you place no value on the other side of it. We think that certainty is very hard to earn honestly, because people's future selves reliably turn out to care. But we are not going to pretend the assumption isn't there. It is the one thing this site cannot prove to you.

Smoking kills, stopping helps

Now you are old enough to weigh the long clock honestly, so here it is without softening.

Smoking kills, stopping helps (cont.)

Cigarettes are the most clearly proven cause of early death in all of medicine. Lifelong smokers die on average about **ten years younger** than people who never smoked, and roughly two in every three lifelong smokers are eventually killed by it. This is not a weak statistical hint; it is one of the most repeated findings in science, shown in different countries, in men and women, across decades, with the biological mechanism traced all the way down to the exact DNA damage. The details and sources are in the evidence below — check them.

But the second half is what turns a grim fact into a decision you control:

Smoking kills, stopping helps (cont.)

- **The risk is not banked. It is largely reversible if you get out early.** Stopping before 40 avoids the large majority of the extra risk; stopping before 30 avoids almost all of it. Even stopping at 60 buys back years. There is no age at which quitting fails to help.
- **Starting is the only part of the story fully in your hands.** Almost all lifelong smokers start before 20. Not starting removes the entire problem at no cost — which is why, if you have made it this far without smoking, you have already made the most valuable move.
- **And if you already smoke, stopping is not a coin-flip of willpower.** Proper support — help from an adviser plus medication — and effective medicines (nicotine replacement, varenicline, cytisine) multiply the odds of success; the numbers are in the evidence below.

Smoking kills, stopping helps (cont.)

On vaping, the honest position is the same one as the younger band, just with the evidence attached: nicotine vaping delivers far fewer of the chemicals made by burning tobacco, and there is high-quality evidence it helps *people who already smoke* to quit — for a smoker, switching is a genuine gain. For a never-smoker none of that applies: it is addiction with no offsetting benefit and no long-term safety record. (The 2019–20 lung-injury scare, EVALI, was traced to vitamin E acetate in illegal cannabis (THC) cartridges, not nicotine vapes — getting that wrong is the fastest way to lose an argument you should win.)

Alcohol: no safe dose

Here is the claim in full, with its honest limits. "No safe dose" packs two separate facts together, and you need both — the scary half on its own would overstate the risk, and people would be right to stop listening.

Alcohol: no safe dose (cont.)

- **The risk starts at the first drink.** There is no low amount where the risk is zero, or where alcohol protects you. The old "J-curve" — the graph that seemed to show light drinkers had healthier hearts than non-drinkers — was a mistake: the "non-drinkers" it compared against included people who had *stopped* drinking because they were already ill, which made not drinking look worse than it is.
- **But the risk from a little is small.** The extra risk from one drink a week is tiny. There is also no *perfectly* safe amount of cycling, or sunlight, or getting into a car. So the honest version keeps the truth and drops the exaggeration: *there is no amount of alcohol that is good for you — there never was — and the risk starts at your first drink and rises with every one after it.*

Alcohol: no safe dose (cont.)

How today affects later: because there is no safe cut-off, risk builds up with the total you drink over your life, rather than waiting until you cross some "safe limit". What matters is the long-run pattern, not any single night — and the clearest example is breast cancer, where risk rises steadily with each daily drink, starting from the first (see the evidence).

Gambling: designed so you lose

The maths needs no statistics to prove it. (Scientists usually use a *p-value* to judge whether a pattern in their data could just be down to chance. Here there is nothing to judge — nobody is estimating anything from data.) The average loss per bet — the product's negative *expected value* — is a **published design setting**, not a research finding. The operator chose it in advance and printed it in the terms. Play long enough and the ups and downs of luck (the *variance*) even out; the edge does not.

Gambling: designed so you lose (cont.)

On top of that sits a second, weaker claim that does need research — that the resulting harm is large and follows the product, not the person — and it is graded honestly in the evidence below: harm is measurably higher for the fastest, most non-stop products (online slots) than for slower ones (sports betting), the same way the harm from a cigarette depends on how much tar it delivers, not on who smokes it.

Gambling: designed so you lose (cont.)

How today affects later: in bank data, the more of their income people spent on gambling, the more likely they were to fall into money trouble — and the very heaviest gamblers were much more likely to die early. Teenagers already gamble. The most protective move comes first — recognising that the machine's edge is fixed and published, and not opening the app in the first place. The claims that gambling *causes* debt, relationship breakdown and suicide are real but tangled up with other causes such as poverty and mental illness; the maths is not tangled up with anything.

Eat food. Mostly plants. Not too much.

Start with the part that is strong, because it is unusually strong for nutrition. **Excess body fat (adiposity) causes type 2 diabetes and heart disease** — Mendelian-randomisation studies, which compare people by genes fixed at birth, show this is cause and effect rather than something else hiding behind it (confounding). The WHO's cancer agency (IARC) judged that excess body fat **causes at least a dozen cancers**. And the risk of dying rises with BMI above the mid-20s. So "why it matters" is not in doubt.

Eat food. Mostly plants. Not too much. (cont.)

The other strong, and under-appreciated, fact is about **reversibility**: the body defends a fat mass once it is reached. After substantial weight loss, the body keeps burning less energy at rest and the weight is usually regained; and obesity tracks powerfully from childhood — a child with obesity is roughly five times as likely to carry it into adulthood, and the great majority of adolescents with obesity stay that way. The practical upshot is stark: **the biggest risk factor for carrying excess weight is having carried it already**. Prevention beats reversal by a wide margin, which is why the leverage is early and in the environment.

Eat food. Mostly plants. Not too much. (cont.)

Two environmental levers on *how much* you eat have unusually clean experimental support: how **processed** the food is (people eat more of it, unprompted) and how **large the portion** is (serve more and people eat more, with no matching rise in fullness). Both are about changing defaults, not out-arguing appetite — a smaller plate and a less-processed cupboard do work that willpower rarely wins.

Eat food. Mostly plants. Not too much. (cont.)

Where the evidence is *weak* is the next question — *which named diet* to prescribe. Michael Pollan's *eat food, mostly plants, not too much* works less as a proven prescription than as a rule about **what to buy**. So the honest position is: the stakes are high and well-evidenced, and the defaults governing how much you eat are worth changing — but no specific named diet is a proven fix, and using smaller portions as a deliberate weight-loss tool meets the same defended-mass wall as everything else, so the effect on intake is far better evidenced than any lasting effect on the scale.

Move more, sit less

Narrowly: physical activity lowers the risk of dying early; the benefit is largest for people moving *out of the least active* group; and the amount that captures most of it is far below the round numbers in circulation.

The more you move, the lower the risk — but the benefit is **front-loaded**: the largest drop in risk comes from leaving the least-active quarter of people, measured with movement trackers (accelerometers) so it isn't inflated by people over-reporting. Getting off the floor matters far more than optimising the top.

Move more, sit less (cont.)

How today affects later: the obvious worry is reverse causation — ill people move less, which would make movement look better than it is. Mendelian-randomisation work, which compares people by genes linked to how active they are, points to a genuine protective effect. Do not oversell the top of the curve, where other differences between very active and less active people muddy the picture; the claim is strongest exactly where it matters most for a population — at the bottom, where the effect is largest and the cause-and-effect evidence best.

Move more, sit less (cont.)

And the benefit is not only on the seventy-year clock. Physical activity is one of the best-evidenced near-term levers on mood, reliably easing symptoms of depression and anxiety over weeks — and a Mendelian-randomisation study supports activity *causing* lower depression risk, since genes are fixed at birth and cannot be explained away by "happier people move more." That near-term effect on how you feel is why this page sits on both clocks.

Sleep like it matters

Narrowly: enough sleep reliably improves mood, attention and emotional control over days to weeks — that part has been tested in experiments — and teenagers' late sleep timing is biology, not laziness. The long-term "sleep saves your life" claim is much weaker, and we do not lean on it.

Sleep like it matters (cont.)

The honest case for sleep is the **two weeks, not the fifty years**. Experiments that cut people's sleep short reliably make attention, mood and emotional control worse; that is the immediately relevant, well-evidenced part. The environmental lever is real too: a later school start produced more sleep, measured with wrist trackers, with better grades and attendance.

Sleep like it matters (cont.)

How today affects later: honestly, the long-term cause-and-effect story is weak — studies linking sleep length to dying early are badly muddled by reverse causation, because illness can both shorten and lengthen sleep, and there will never be an experiment on lifetime sleep. So the durable move is to treat sleep as a daily lever on how you perform and feel, and protect its conditions (light, timing, a phone that isn't in the bed), rather than to frame it as a distant survival threat.

Slip, slop, slap

Narrowly: regular sunscreen use reduces skin cancer, and — unusually for an everyday habit — this rests on a genuine randomised controlled trial rather than just observation. The line itself is a model of how to write health advice: three verbs, an action for each, no threat and no blame.

The Nambour trial randomly split a whole community's volunteers into daily versus use-when-you-like sunscreen, and ten years *after the trial stopped* the daily-sunscreen group had roughly half the melanomas. That the benefit outlasted the trial is the striking part, and it is why this is the strongest study design on the site.

Slip, slop, slap (cont.)

How today affects later: melanoma is already among the commonest cancers in your age group, so this is one of the very few items where the ten-year and seventy-year clocks agree; and because damage builds up, the protective habit pays both now and later. Be honest about the limits, though — it is one trial, one fair-skinned, high-sunshine population, melanoma was not the main thing the trial was set up to measure (a *secondary endpoint*), and the invasive-melanoma result rests on very small numbers. Even the best study here does not deliver certainty.

Change the rules, not just yourself

This page is the logical endpoint of everything else here — and it points at your ballot. If how much people smoke, drink, eat and move is powerfully shaped by their environment, then the highest-leverage health action is often to **change the environment**, and for a whole population that means policy, decided collectively.

Change the rules, not just yourself (cont.)

The evidence is strongest exactly where the measure changes the environment and the study design is clean. Scotland's indoor smoking ban was followed by a sharp, repeated fall in heart-attack admissions, most of it in non-smokers who had merely been breathing the air. Its minimum unit price for alcohol was followed by a marked reduction in deaths caused by alcohol, concentrated in the most deprived communities. The UK's soft-drinks levy made manufacturers take the sugar out of their drinks, and was followed by a targeted reduction in childhood obesity in the poorest areas. (The precise figures, with sources, are in the evidence below.) Be honest about the weaker case, too: **active-travel infrastructure** — cycle paths, safe routes — is suggestive, not proven.

Change the rules, not just yourself (cont.)

Put those together and an honest conclusion follows: the biggest population health improvements have generally come from changing the shared environment, not from urging individuals to try harder — so **engaging as a citizen, and voting for representatives who prioritise health-promoting environments over ill-health-promoting ones, is itself a high-leverage, evidence-based health act.** This is a reference, not a campaign, so it stops there: the evidence for *what works* is above; *which* measures are worth their trade-offs, and *who* best represents them, is a question of values that is yours to answer, not ours.

Five things to remember

- 1. Don't smoke — and if you do, stop, the earlier the better.**
- 2. Move more, starting from the bottom.**
- 3. Don't let excess weight arrive.**
- 4. With alcohol, less is better — all the way down to none.**
- 5. Use your voice and your vote.**

Trust, but verify.

Full evidence — with the study designs, the effect sizes, and the honest weaknesses — is on the site.

Information is power.